

Zachary Benson

Senior full stack engineer (Java, Spring Boot, React, AWS, TypeScript, Angular, Android)

Portland, Oregon, US

+1 (971) 209-7282

react1205@gmail.com

<https://www.linkedin.com/in/zachary-benson-6578a33b5/>

PROFESSIONAL SUMMARY

Versatile and results-oriented Senior Full Stack Engineer with 8+ years of experience architecting and delivering high-performance, scalable web applications using Java, Spring, Spring Boot, React, Node.js, Vue.js, Angular, Micronaut, C# and TypeScript.

Skilled in both frontend and backend development, with deep expertise in building responsive UIs, secure APIs, and modular, maintainable architectures. Proficient in integrating RESTful and GraphQL APIs, optimizing performance through lazy loading, code splitting, and SSR/SSG strategies. Experienced with cloud infrastructure using AWS, Docker, and Kubernetes, and implementing robust CI/CD pipelines for efficient deployment.

Strong background in Agile/Scrum teams, leading full project lifecycles, mentoring developers, and enforcing best practices in code quality, testing, and documentation. Passionate about solving real-world problems, aligning technology with business goals, and delivering impactful, user-focused digital experiences across the stack.

EDUCATION

University of Florida - Bachelor's Degree in computer Engineering, 2014 - June 2018

TECHNICAL SKILLS

Backend & Microservices Architecture: Advanced expertise in Java, Kotlin, and Node.js (TypeScript) for building resilient, scalable systems. Proficient in Spring Boot, Spring Security, and Express.js, with deep experience in Microservices, BFF, Event-Driven Architecture, and Domain-Driven Design (DDD).

Mobile Engineering (Android Native): Specialist in Android development using Kotlin and Java. Expert in modern UI frameworks like Jetpack Compose and Material Design, utilizing MVVM/MVI and Clean Architecture to build robust, testable applications powered by Coroutines, Flow, and Hilt/Dagger.

API Design & Communication: Architected and implemented high-performance communication layers using REST, GraphQL, gRPC, and WebSockets. Skilled in managing asynchronous messaging and decoupling services via Kafka, RabbitMQ, and Google Cloud Pub/Sub.

Frontend & Interactive UIs: Proficient in building responsive, component-based web applications using React, Next.js, and Angular. Experienced in state management via Redux Toolkit and RxJS, ensuring top-tier UX with Tailwind CSS and Material UI.

Cloud Infrastructure & DevOps: Deep hands-on experience with AWS (Lambda, EC2, EKS, Fargate) and GCP (GKE, Apigee, BigQuery). Skilled in Infrastructure as Code (IaC) using Terraform and automating full-cycle delivery through Jenkins, GitHub Actions, and GitLab CI with ArgoCD for Kubernetes.

Database Systems & Persistence: Expert in designing and optimizing relational schemas (PostgreSQL, MySQL) and NoSQL solutions (MongoDB, DynamoDB). Proficient in high-performance caching with Redis and implementing advanced search capabilities via Elasticsearch and OpenSearch.

Security & Identity Management: Implemented enterprise-grade security protocols including OAuth 2.0, JWT, and OpenID Connect. Experienced in securing cloud environments using AWS Cognito, GCP Secret Manager, and RBAC patterns.

Testing, Quality & Observability: Committed to TDD/BDD practices using JUnit, Mockito, Espresso, and Cypress. Proactive in system health monitoring and debugging using Grafana, Datadog, Sentry, and Prometheus to ensure 99.9% uptime.

Agile Leadership & Collaboration: Strong background in Agile/Scrum methodologies, utilizing Jira and Confluence for streamlined project tracking. Experienced in leading code reviews and architectural discussions within globally distributed engineering teams.

WORK EXPERIENCE

Deloitte (Remote)- Java Full Stack Engineer

Jul 2024 – Present

- Architected and delivered a high-performance, **cloud-native e-commerce** platform using the Java stack (**Spring Boot**, **PostgreSQL**, **React**) within a **Microservices** architecture, ensuring scalability, modularity, and independent deployments via **Spring Cloud**.
- Led the design and architecture of **RESTful** and **GraphQL** APIs using Java and **Netflix DGS/Spring GraphQL**, establishing consistent patterns for versioning, error handling, and authentication (**Spring Security**, **JWT**, **OAuth2/OIDC**).
- Created a centralized API Gateway pattern with rate limiting and request validation using **Spring Cloud Gateway** and **Redis**, significantly improving latency and throughput across distributed services.
- Designed and built a highly scalable **React** component architecture using **TypeScript**, following atomic design principles and Storybook-driven development, supporting reusable, testable, and responsive UI components.
- Implemented **Next.js SSR/SSG** strategies, dynamic imports, and lazy loading to enhance frontend performance, **Core Web Vitals**, and **SEO** readiness for global **e-commerce traffic**.
- Defined and maintained backend modules with Spring Boot, applying **Domain-Driven Design (DDD)**, dependency injection, and **Spring Data JPA (Hibernate)** to improve maintainability and separation of concerns.
- Integrated secure payment workflows with **Stripe** and **PayPal**, ensuring PCI-DSS compliance, and built extensible Spring Interceptors and middleware for webhook handling and fraud detection.
- Built and deployed cloud infrastructure using AWS (App Runner, S3, RDS, Lambda) and Terraform for Infrastructure as Code (IaC), supporting high availability and disaster recovery.
- Automated full CI/CD pipelines using GitHub Actions and Jenkins, utilizing Docker, Helm, and Amazon EKS (Kubernetes) to enable canary releases and environment parity.
- Implemented full-stack observability using **Micrometer**, **Prometheus**, **Grafana**, and **ELK Stack**, establishing standardized dashboards and alerts for deep production insights.
- Applied **Test-Driven Development (TDD)** using JUnit 5, **Mockito**, **Cypress**, and **React Testing Library**, ensuring robust test coverage across unit, integration, and **E2E** flows.
- Collaborated with designers to translate Figma wireframes into responsive, accessible, and scalable UI systems, promoting consistent branding across platforms.
- Authored technical documentation, API specifications (**OpenAPI/Swagger**), and **DevOps** runbooks to support ongoing development and team knowledge sharing.
- Mentored frontend and backend engineers on Clean Architecture, code quality, and Agile/Scrum delivery practices, contributing to a high-performing, cross-functional team.

General Motors (Florida) – Java Full Stack Engineer

Oct 2022 - May 2024

- Architected and developed a real-time vehicle tracking platform using the **Java stack (Spring Boot, PostgreSQL, Angular)**, supporting thousands of concurrent users and delivering accurate live telemetry and fleet performance insights.
- Designed and deployed cloud-native infrastructure, leveraging **Amazon S3, AWS Lambda, API Gateway, and Amazon Cognito (or Spring Security OAuth2)**, ensuring secure authentication and highly available backend services.
- Engineered real-time data streaming systems using **Spring WebFlux** and **WebSockets**, enabling instant map updates, route tracking, and telemetry broadcasting with low-latency data handling in a distributed reactive environment.
- Developed secure authentication and authorization layers using **Spring Security, OAuth2, JWT**, and **Multi-Factor Authentication (MFA)**, enhancing system resilience and data protection.
- Built scalable, reusable UI components and Directives with **Angular + TypeScript**, implementing advanced state management patterns with **NgRx** and Services to ensure consistent and performant frontend behavior.
- Designed and implemented **RESTful APIs** and **GraphQL** endpoints (**Netflix DGS**) for modular data consumption, aligning backend services with scalable microservice communication models using Java 17/21.
- Optimized database performance using advanced indexing, partitioning, execution plan analysis, and Spring Data JPA (Hibernate), improving data retrieval speeds and reducing query load under high traffic.
- Implemented **CI/CD** pipelines using Jenkins, **GitLab CI**, and Docker, ensuring automated testing, build, and zero-downtime deployment via **Kubernetes (EKS)** across staging and production environments.
- Enabled infrastructure-as-code (IaC) practices using Terraform and **AWS CloudFormation**, standardizing environment provisioning and improving deployment consistency.
- Led development of interactive dashboards using custom charting libraries and Map APIs, utilizing **Angular's** change detection strategies and **RxJS** Observables for high-frequency visual analytics.
- Applied performance tuning techniques such as **Angular Route Guard lazy loading, OnPush** change detection, and distributed caching (**Redis**), significantly reducing initial load times and improving responsiveness.
- Authored and maintained comprehensive technical documentation, API references, architectural diagrams, and operational guides for development and **DevOps** teams.
- Championed **Agile** and **Scrum** methodologies, conducting sprint planning and code reviews while mentoring junior engineers in **SOLID** principles, **Clean Architecture**, and **Java best practices (Effective Java)**.
- Conducted extensive automated and manual testing using **JUnit 5, Mockito, Jasmine**, and **Karma**, ensuring robust **QA** coverage across backend and frontend services.

Zillow (Remote) – Java Full Stack Engineer

Mar 2020 - Sep 2022

- Designed and developed enterprise-level **real estate** platforms using the **Java stack (Spring Boot, PostgreSQL, Angular)**, delivering intuitive property search, listings, and management systems.
- **Architected** and deployed **cloud-native infrastructure**, leveraging **Amazon S3, AWS Lambda** (or **Google Cloud Functions**), and **API Gateway**, ensuring fault-tolerant, scalable **backend** services and efficient **serverless** data workflows.
- Developed advanced frontend features such as interactive map visualizations, dynamic filters, and real-time property listings, enhancing UX across web and mobile interfaces using **Angular** and **TypeScript**.
- Built and secured **RESTful APIs** for real-time data access and third-party integrations, including payment gateways, geolocation services, and MLS property data providers using Java 17/21.
- Implemented **SSR** (Server-Side Rendering) with **Angular** Universal and Spring Boot, incorporating lazy loading and intelligent caching strategies to significantly improve **SEO** and application performance.
- Optimized complex client-side state using **NgRx** and **RxJS**, enabling responsive UI behavior and consistent user session experiences across enterprise property dashboards.
- Integrated **TypeScript** and enforced modern development standards, improving maintainability, scalability, and type safety across the full **Spring Boot/Angular** stack.
- Collaborated with product owners and **UI/UX** teams under **Agile** and **Scrum** methodologies to prioritize backlogs, estimate workloads, and meet sprint deliverables within **Jira**.
- Led code reviews and provided technical mentorship on **Clean Architecture**, **SOLID** principles, and efficient component design patterns for both frontend and backend teams.
- Authored detailed system documentation, architectural flow diagrams, and deployment guides to support onboarding, scalability, and cross-team alignment.
- Automated testing workflows using **JUnit 5, Mockito, Jasmine**, and **Karma**, ensuring high code coverage, regression protection, and reliable production releases.
- Initiated performance audits and implemented database indexing, query optimization via Spring Data JPA (**Hibernate**), and **CDN** integration, reducing latency across key workflows.

TePe USA (California) – Full Stack Engineer

Sep 2018 – Feb 2020

- Architected high-scale enterprise healthcare platforms using **Angular and TypeScript**, focusing on reusable UI components, **modular design patterns**, and **Angular Feature Modules**.
- Leveraged **Angular Material** for consistent CSS styling; engineered complex Menus and Drawer components using TypeScript for strict type safety and **reactive component communication**.
- Integrated the **html2canvas** package within the Angular ecosystem to provide advanced client-side control over image rendering and screen capture functionality.
- Implemented robust identity management and secure user authentication flows utilizing **Spring Security** and **Auth0**.
- Engineered high-performance backend services using **Java and Spring Boot (Web API)**, adhering to **Domain-Driven Design (DDD)** and **SOLID** principles for clean separation of concerns.
- Managed relational data schemas with **PostgreSQL** (or MySQL) and utilized **Spring Data JPA (Hibernate)** for database control and object-relational mapping (ORM).
- Streamlined data transfer using **MapStruct** (or ModelMapper) for DTO mapping; integrated **SpringDoc OpenAPI** to generate interactive **Swagger UI** documentation.
- Conducted rigorous code quality audits and performance profiling using **SonarQube** and browser profiling tools to optimize frontend delivery and backend efficiency.
- Managed the full deployment lifecycle for frontend and backend projects hosted on **DigitalOcean**, utilizing droplets and **Docker** for high availability.
- Applied **TDD principles** using **JUnit 5**, **Mockito**, and **Jasmine/Karma** to ensure high code coverage for critical business logic across the full stack